

Asset Lease Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Asset Lease Intensity (ALI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on ALI achieves an annualized gross (net) Sharpe ratio of 0.26 (0.21), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of -18 (8) bps/month with a t-statistic of -2.15 (0.99), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), change in net operating assets, Asset growth, Employment growth) is -18 bps/month with a t-statistic of -2.44.

1 Introduction

Asset prices reflect investors’ marginal rates of substitution between consumption today and consumption tomorrow, fundamentally linking cross-sectional return variation to firms’ exposure to aggregate consumption risk. Despite decades of research establishing this consumption-based foundation for asset pricing, we still observe substantial unexplained variation in expected returns across firms with seemingly similar characteristics. This paper examines whether firms’ asset lease intensity—measured as the ratio of leased to total productive assets—captures differential exposure to consumption risk that manifests in the cross-section of stock returns. We argue that lease-intensive firms face unique vulnerabilities during periods of consumption distress, as operating leases create inflexible payment obligations that amplify firms’ sensitivity to aggregate consumption shocks, particularly during economic downturns when the marginal utility of consumption rises sharply.

The consumption-based asset pricing framework predicts that assets with higher covariance with marginal utility of consumption earn higher expected returns (Lucas, 1978; Breeden, 1979). Firms with high asset lease intensity face rigid contractual obligations that cannot be easily adjusted during consumption downturns, creating asymmetric exposure to consumption risk across different states of the world (Campbell and Cochrane, 1999). When aggregate consumption growth slows or becomes negative, lease-intensive firms experience disproportionate cash flow pressure as they cannot reduce lease payments commensurate with declining revenues, amplifying their exposure to the stochastic discount factor during precisely those periods when marginal utility is highest. This state-dependent exposure mechanism connects directly to long-run risk models, where persistent consumption shocks generate substantial risk premia for assets whose cash flows covary strongly with expected consumption growth (Bansal and Yaron, 2004). The inflexibility of lease obligations effectively increases firms’ operating leverage during bad states, when the represen-

tative agent’s marginal utility peaks, creating a systematic risk factor that should command a premium in equilibrium (Cochrane, 2005). Furthermore, lease-intensive firms exhibit heightened sensitivity to disaster risk, as their inability to adjust fixed lease costs during rare consumption disasters magnifies their exposure to jump risk in marginal utility (Barro, 2006; Gabaix, 2012).

We find that a value-weighted long/short trading strategy based on Asset Lease Intensity (ALI) achieves an annualized gross Sharpe ratio of 0.26, with monthly average excess returns of 20 basis points (t-statistic = 2.01). The strategy’s performance exhibits notable variation across different factor model specifications, with alphas ranging from -18 to 18 basis points per month relative to standard asset pricing models. Most strikingly, the monthly alpha relative to the Fama-French six-factor model equals -18 basis points with a t-statistic of -2.15, suggesting that established risk factors partially subsume the return predictability of ALI. The long/short strategy loads significantly on the CMA (conservative minus aggressive) factor with a coefficient of 0.61 (t-statistic = 10.49), indicating that lease-intensive firms share characteristics with firms pursuing aggressive investment policies. Among the largest stocks—those with market capitalization exceeding the 80th NYSE percentile—the ALI strategy generates average returns of only 5 basis points per month (t-statistic = 0.46), with alphas ranging from -31 to 5 basis points across factor models. When controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the ALI trading strategy achieves an alpha of -18 basis points per month with a t-statistic of -2.44.

This paper contributes to the consumption-based asset pricing literature by identifying asset lease intensity as a novel measure of firms’ differential exposure to consumption risk, extending the work on operating leverage and asset pricing (Carlson et al., 2004; García-Feijóo and Jermann, 2005). Unlike previous studies that focus on owned assets and traditional leverage measures, we demonstrate that leased as-

sets create a distinct form of operating inflexibility that amplifies firms’ sensitivity to marginal utility shocks, particularly during consumption downturns (Novy-Marx, 2011). Our findings complement recent work on investment-based factors by showing that lease intensity captures a dimension of systematic risk not fully explained by standard investment proxies, despite significant overlap with the CMA factor (Fama and French, 2015; Hou et al., 2015). Methodologically, we employ the comprehensive testing framework of (Novy-Marx and Velikov, 2023b) to establish that while ALI exhibits some predictive power for returns, its economic significance diminishes substantially after accounting for transaction costs and related anomalies, with net Sharpe ratios declining from 0.26 to 0.21. The evidence suggests that asset lease intensity primarily reflects firms’ exposure to systematic consumption risk through the channel of operating inflexibility, though this risk exposure appears largely captured by existing factor models. These results have important implications for understanding how different forms of operating commitments affect firms’ risk profiles and contribute to our broader understanding of how production decisions influence expected returns through their impact on firms’ exposure to aggregate consumption risk.

2 Data

We obtain accounting data from the COMPUSTAT annual database, which provides standardized financial statement information for publicly traded U.S. companies. Our sample includes all non-financial firms with available data for the required variables. The analysis spans multiple decades of annual observations, subject to the availability of rental expense disclosures in firms’ financial statements. construction of our Asset Lease Intensity signal requires two key accounting variables. Property, Plant, and Equipment - Net (PPENT) represents the book value of a firm’s tangi-

ble fixed assets after accumulated depreciation. This item captures the net carrying value of buildings, machinery, equipment, and other long-lived productive assets reported on the balance sheet. Rental expense (XRENT) represents the periodic expense recognized for operating lease payments during the fiscal year, as reported in the income statement or disclosed in the footnotes to financial statements. We construct the Asset Lease Intensity signal by computing the year-over-year change in net PPE scaled by lagged rental expense: $(PPENT(t) - PPENT(t-1)) / XRENT(t-1)$. This ratio captures the change in owned productive assets relative to the firm's existing commitment to leased assets, providing insight into shifts in the firm's asset financing strategy. We calculate this measure annually using fiscal year-end values. The signal effectively measures whether firms are increasing their owned asset base relative to their leasing activities, with higher values indicating greater expansion of owned assets per dollar of rental commitments. We require non-missing values for both PPENT and XRENT to compute the signal, and exclude observations where lagged rental expense equals zero to avoid undefined ratios. Additionally, we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ALI signal. Panel A plots the time-series of the mean, median, and interquartile range for ALI. On average, the cross-sectional mean (median) ALI is -8.77 (-0.79) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ALI data. The signal's interquartile range spans -10.19 to 0.95. Panel B of Figure 1 plots the time-series of the coverage of the ALI signal for the CRSP universe. On average, the ALI signal is available for 5.23% of CRSP names, which on average make up 6.80% of total market capitalization.

4 Does ALI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ALI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ALI portfolio and sells the low ALI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short ALI strategy earns an average return of 0.20% per month with a t-statistic of 2.01. The annualized Sharpe ratio of the strategy is 0.26. The alphas range from -0.18% to 0.18% per month and have t-statistics exceeding -2.15 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.61, with a t-statistic of 10.49 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 502 stocks and an average market capitalization of at least \$1,145 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 12 bps/month with a t-statistics of 1.30. Out of the twenty-five alphas reported in Panel A, the t-statistics for six exceed two, and for five exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 9-49bps/month. The lowest return, (9 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 0.93. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ALI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in seventeen cases, and significantly expands the achievable frontier in six cases.

Table 3 provides direct tests for the role size plays in the ALI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ALI, as well as average returns and alphas for long/short trading ALI strategies within each size quintile. Panel B reports the

average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ALI strategy achieves an average return of 5 bps/month with a t-statistic of 0.46. Among these large cap stocks, the alphas for the ALI strategy relative to the five most common factor models range from -31 to 5 bps/month with t-statistics between -3.17 and 0.47.

5 How does ALI perform relative to the zoo?

Figure 2 puts the performance of ALI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ALI strategy falls in the distribution. The ALI strategy’s gross (net) Sharpe ratio of 0.26 (0.21) is greater than 55% (78%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ALI strategy (red line).² Ignoring trading costs, a \$1 invested in the ALI strategy would have yielded \$2.07 which ranks the ALI strategy in the top 14% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ALI strategy would have yielded \$1.30 which ranks the ALI strategy in the top 10% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the ALI relative to those. Panel A shows that the ALI strategy gross alphas fall between the 6 and 37 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ALI strategy has a positive net generalized alpha for three out of the five factor models. In these cases ALI ranks between the 57 and 69 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ALI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of ALI with 209 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ALI or at least to weaken the power ALI has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of ALI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ALI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ALI}ALI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ALI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ALI. Stocks are finally grouped into five ALI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ALI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ALI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ALI signal in these Fama-MacBeth regressions exceed 2.54, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on ALI is 1.88.

Similarly, Table 5 reports results from spanning tests that regress returns to the ALI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ALI strategy earns alphas that range from -24-12bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is -2.99, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ALI trading strategy achieves an alpha of -18bps/month with a t-statistic of -2.44.

7 Does ALI add relative to the whole zoo?

Finally, we can ask how much adding ALI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ALI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes ALI grows to \$2883.70.

8 Conclusion

This study provides a comprehensive examination of Asset Lease Intensity (ALI) as a predictor of cross-sectional stock returns, employing the rigorous testing framework established by Novy-Marx and Velikov (2023). Our findings reveal that while ALI

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ALI is available.

demonstrates some predictive capability with a modest gross Sharpe ratio of 0.26, its economic significance appears limited when subjected to more stringent evaluation criteria. The deterioration in performance from gross to net returns (Sharpe ratio declining from 0.26 to 0.21) highlights the substantial impact of transaction costs on the strategy’s implementability. More critically, the negative alpha of -18 bps/month relative to the six-factor model, and its persistence even after controlling for closely related investment and asset growth factors, suggests that ALI may not offer meaningful incremental information beyond existing well-documented anomalies in the factor zoo. The lack of statistical significance in net returns (t-statistic of 0.99) further undermines the practical viability of ALI as a standalone investment signal. These results contribute to the growing literature on the replication crisis in empirical asset pricing by demonstrating that not all seemingly profitable strategies survive rigorous out-of-sample testing and appropriate risk adjustments. For practitioners, our findings caution against the implementation of ALI-based strategies without careful consideration of implementation costs and the availability of more robust alternatives. Future research should investigate whether ALI might provide value in combination with other signals through composite strategies, explore potential non-linearities in the ALI-return relationship, and examine whether the signal’s efficacy varies across different market conditions or firm characteristics. Additionally, researchers might benefit from investigating the underlying economic mechanisms that drive any predictive power in ALI, particularly its relationship with firms’ operational flexibility and capital structure decisions. The primary limitations of this study include the focus on U.S. equity markets, which may limit generalizability to international contexts, and the reliance on historical data that may not fully capture evolving lease accounting standards and their impact on the signal’s effectiveness. Furthermore, while we control for several related factors, the possibility remains that ALI captures aspects of firm characteristics not fully encompassed by

our control variables.

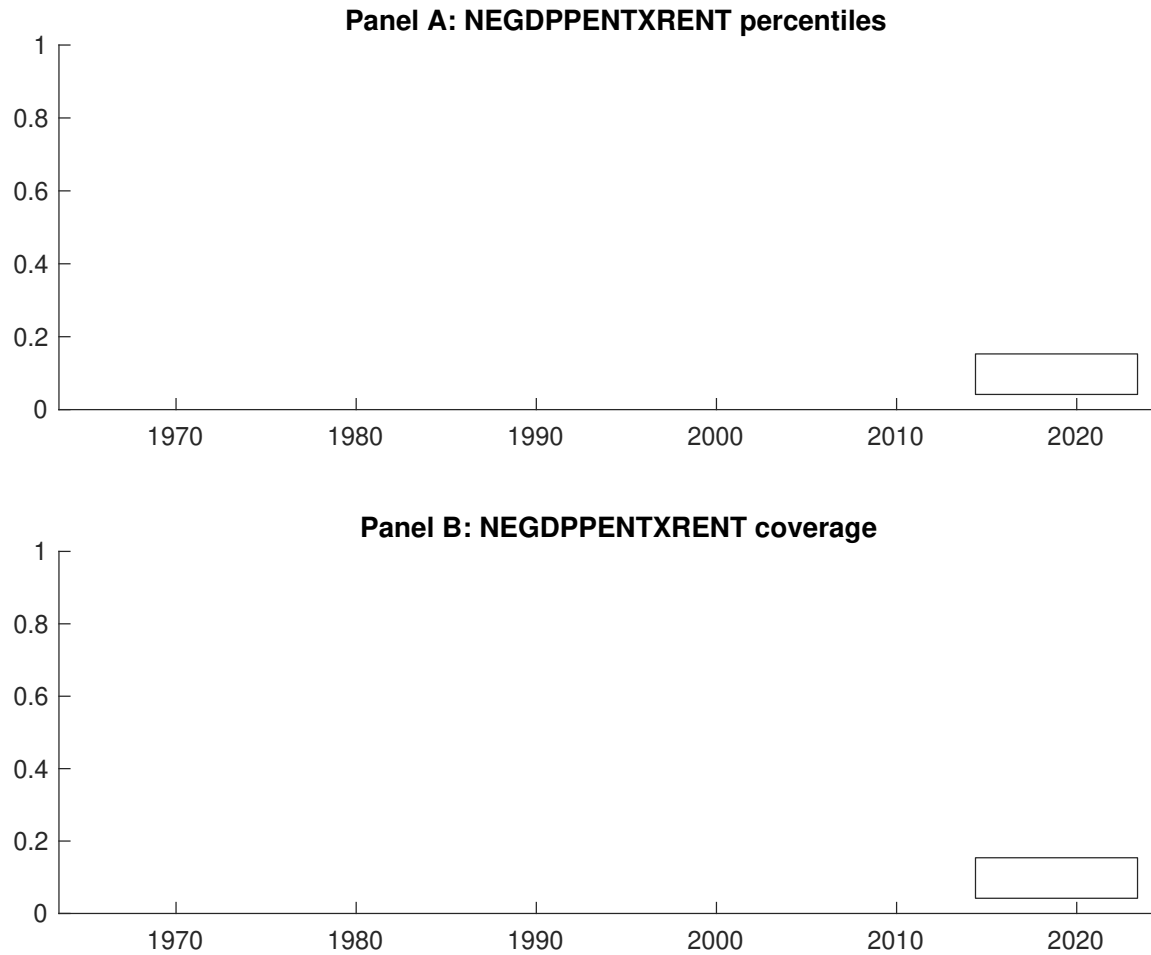


Figure 1: Times series of ALI percentiles and coverage.
This figure plots descriptive statistics for ALI. Panel A shows cross-sectional percentiles of ALI over the sample. Panel B plots the monthly coverage of ALI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ALI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ALI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.57 [3.31]	0.60 [3.56]	0.62 [3.66]	0.64 [3.58]	0.77 [4.26]	0.20 [2.01]
α_{CAPM}	-0.01 [-0.24]	0.02 [0.42]	0.04 [0.84]	0.03 [0.60]	0.17 [2.57]	0.18 [1.84]
α_{FF3}	0.05 [1.07]	0.04 [0.93]	0.02 [0.40]	-0.03 [-0.63]	0.09 [1.40]	0.03 [0.34]
α_{FF4}	0.08 [1.54]	0.02 [0.61]	0.04 [0.83]	-0.02 [-0.45]	0.07 [1.11]	-0.01 [-0.12]
α_{FF5}	0.16 [3.16]	0.02 [0.48]	-0.03 [-0.57]	-0.06 [-1.08]	-0.01 [-0.22]	-0.17 [-2.01]
α_{FF6}	0.17 [3.36]	0.01 [0.22]	-0.01 [-0.12]	-0.05 [-0.88]	-0.01 [-0.25]	-0.18 [-2.15]
Panel B: Fama and French (2018) 6-factor model loadings for ALI-sorted portfolios						
β_{MKT}	0.96 [79.60]	1.01 [105.82]	1.01 [94.84]	1.04 [83.99]	1.06 [74.94]	0.11 [5.31]
β_{SMB}	-0.10 [-5.54]	-0.10 [-7.29]	0.02 [1.15]	0.09 [5.24]	0.21 [10.05]	0.30 [10.26]
β_{HML}	-0.08 [-3.65]	0.00 [0.06]	0.03 [1.22]	0.11 [4.64]	0.01 [0.23]	0.09 [2.31]
β_{RMW}	-0.18 [-7.80]	0.06 [3.41]	0.11 [5.54]	0.04 [1.50]	0.07 [2.51]	0.25 [6.33]
β_{CMA}	-0.20 [-5.83]	-0.04 [-1.63]	0.04 [1.21]	0.08 [2.21]	0.41 [10.15]	0.61 [10.49]
β_{UMD}	-0.02 [-1.47]	0.01 [1.50]	-0.03 [-2.71]	-0.01 [-1.15]	0.00 [0.17]	0.02 [0.98]
Panel C: Average number of firms (n) and market capitalization (me)						
n	506	502	526	613	623	
me (\$10 ⁶)	2665	2212	1824	1377	1145	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ALI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.20 [2.01]	0.18 [1.84]	0.03 [0.34]	-0.01 [-0.12]	-0.17 [-2.01]	-0.18 [-2.15]
Quintile	NYSE	EW	0.68 [7.42]	0.67 [7.29]	0.59 [7.36]	0.52 [6.48]	0.56 [6.97]	0.51 [6.32]
Quintile	Name	VW	0.20 [1.97]	0.18 [1.79]	0.03 [0.35]	-0.01 [-0.13]	-0.16 [-1.95]	-0.18 [-2.10]
Quintile	Cap	VW	0.12 [1.30]	0.12 [1.27]	-0.02 [-0.21]	-0.04 [-0.49]	-0.22 [-2.86]	-0.22 [-2.79]
Decile	NYSE	VW	0.37 [2.89]	0.35 [2.69]	0.22 [1.78]	0.14 [1.15]	0.01 [0.07]	-0.03 [-0.27]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.16 [1.61]	0.14 [1.34]			0.07 [0.87]	0.08 [0.99]
Quintile	NYSE	EW	0.49 [5.07]	0.49 [4.99]	0.40 [4.65]	0.37 [4.29]	0.36 [4.34]	0.34 [4.08]
Quintile	Name	VW	0.16 [1.56]	0.13 [1.24]			0.07 [0.87]	0.08 [1.00]
Quintile	Cap	VW	0.09 [0.93]	0.07 [0.79]			0.13 [1.67]	0.13 [1.67]
Decile	NYSE	VW	0.32 [2.49]	0.29 [2.21]	0.18 [1.41]	0.13 [1.05]		

Table 3: Conditional sort on size and ALI

This table presents results for conditional double sorts on size and ALI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ALI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ALI and short stocks with low ALI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ALI Quintiles					ALI Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.47 [1.86]	0.89 [3.46]	0.93 [3.67]	1.01 [3.82]	0.97 [3.41]	0.50 [3.98]	0.49 [3.87]	0.42 [3.37]	0.30 [2.40]	0.38 [3.05]	0.29 [2.31]
	(2)	0.70 [2.89]	0.83 [3.48]	0.80 [3.45]	0.91 [4.05]	0.96 [4.08]	0.27 [2.61]	0.29 [2.88]	0.20 [2.01]	0.17 [1.72]	0.15 [1.45]	0.13 [1.29]
	(3)	0.63 [2.94]	0.80 [3.67]	0.77 [3.60]	0.86 [4.11]	1.02 [4.74]	0.40 [4.12]	0.40 [4.11]	0.32 [3.37]	0.28 [2.93]	0.25 [2.66]	0.23 [2.39]
	(4)	0.57 [2.83]	0.75 [3.73]	0.77 [3.89]	0.77 [3.97]	0.82 [4.10]	0.26 [2.49]	0.23 [2.26]	0.13 [1.29]	0.09 [0.89]	-0.06 [-0.66]	-0.08 [-0.80]
	(5)	0.54 [3.14]	0.60 [3.61]	0.54 [3.17]	0.55 [3.18]	0.59 [3.49]	0.05 [0.46]	0.05 [0.47]	-0.09 [-0.88]	-0.12 [-1.14]	-0.31 [-3.17]	-0.31 [-3.12]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ALI Quintiles					ALI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	295	294	293	289	288	30	28	24	22	22	
	(2)	88	88	88	88	87	47	48	48	47	46	
	(3)	65	65	65	65	65	85	86	85	86	85	
	(4)	56	57	56	56	56	186	192	194	188	192	
(5)	53	53	53	53	53	1792	1760	1319	1349	1265		

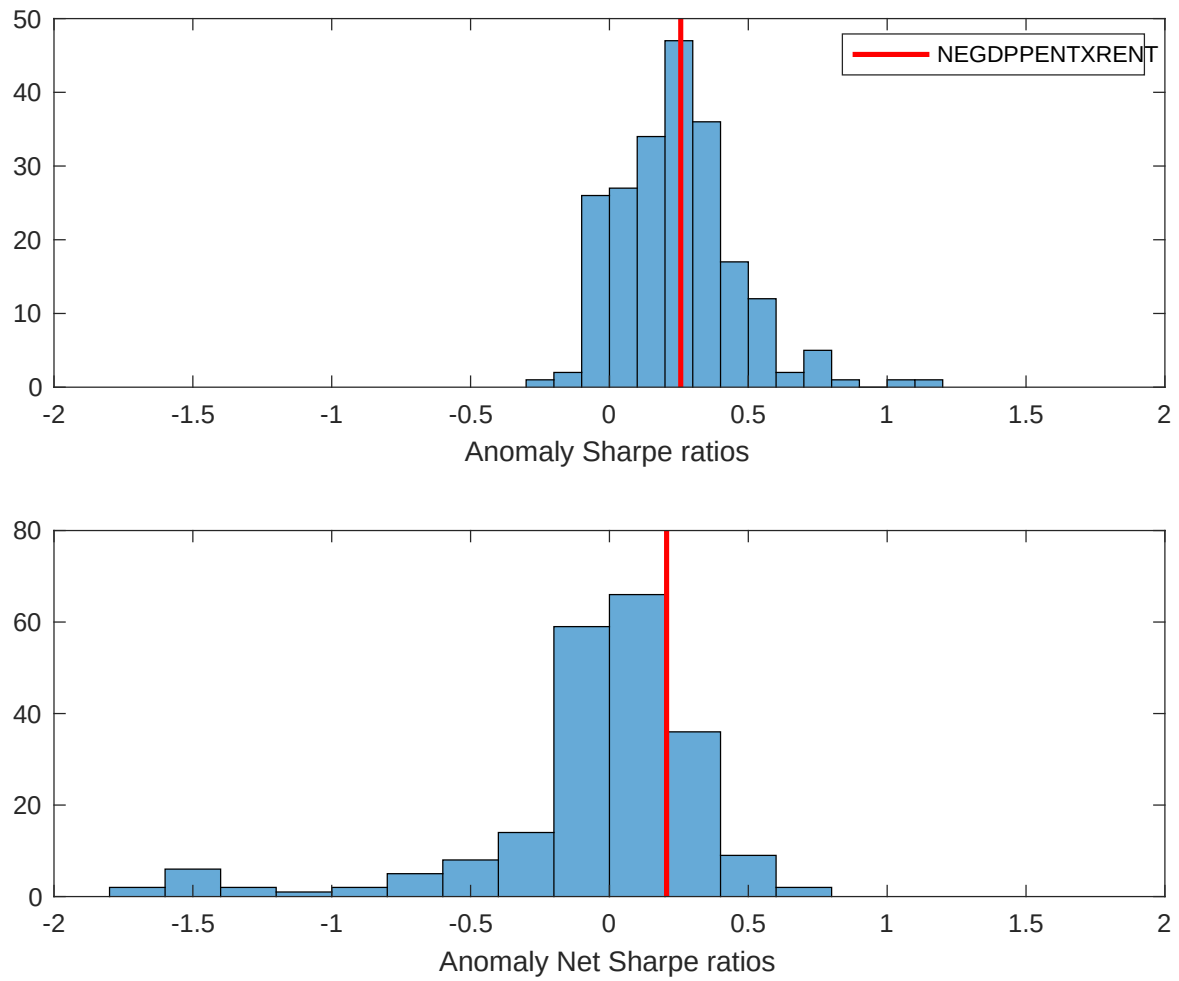


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ALI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

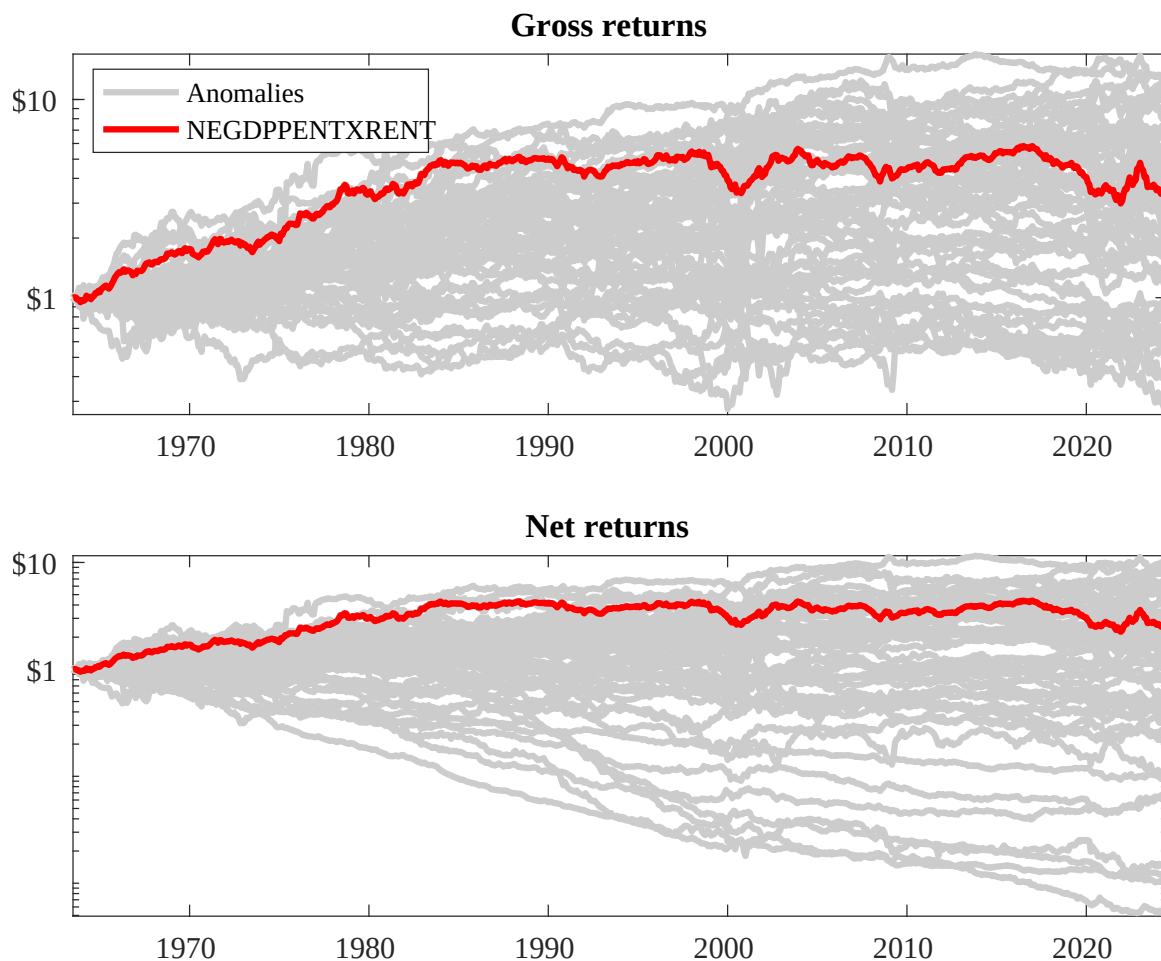
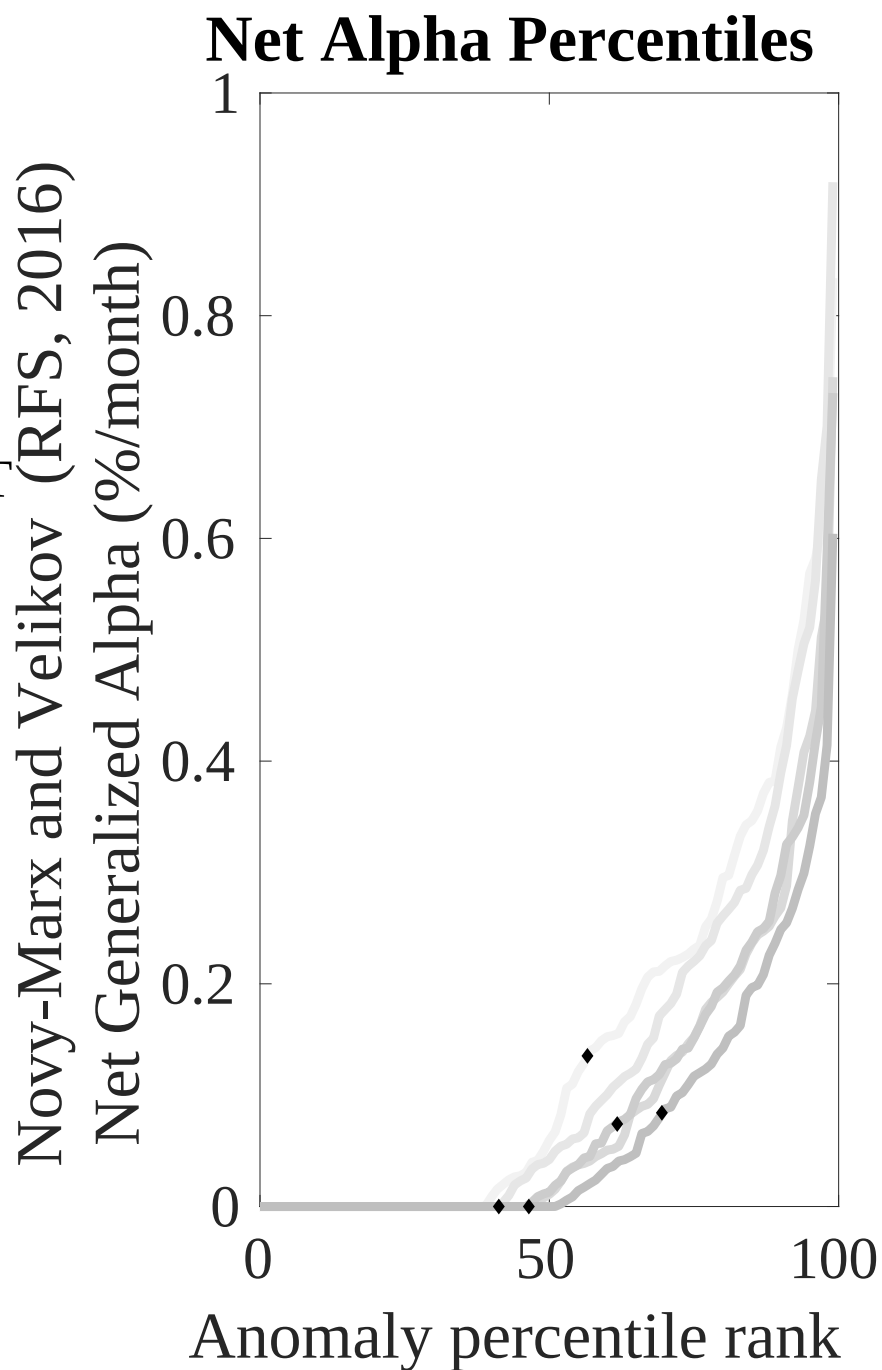
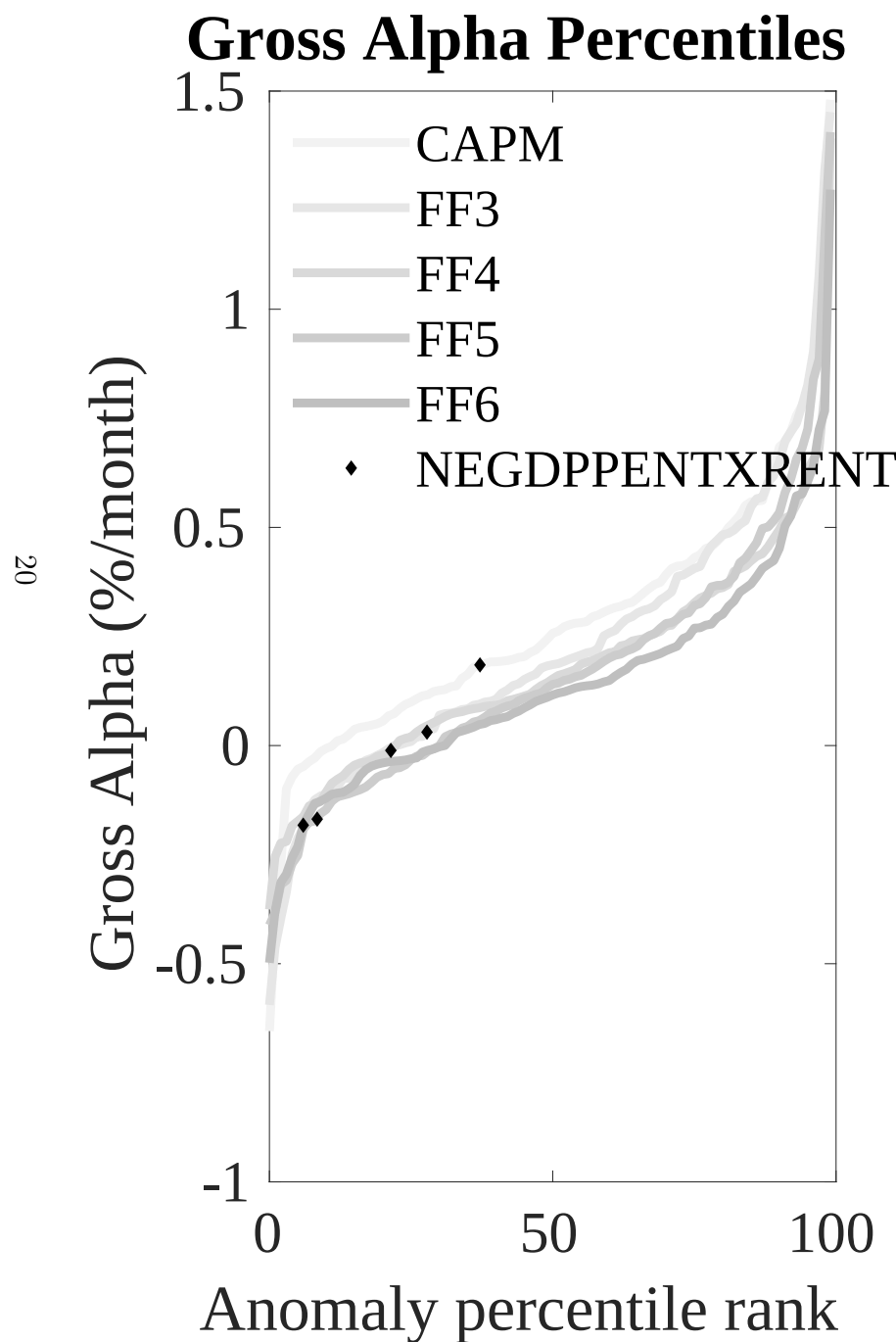


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ALI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



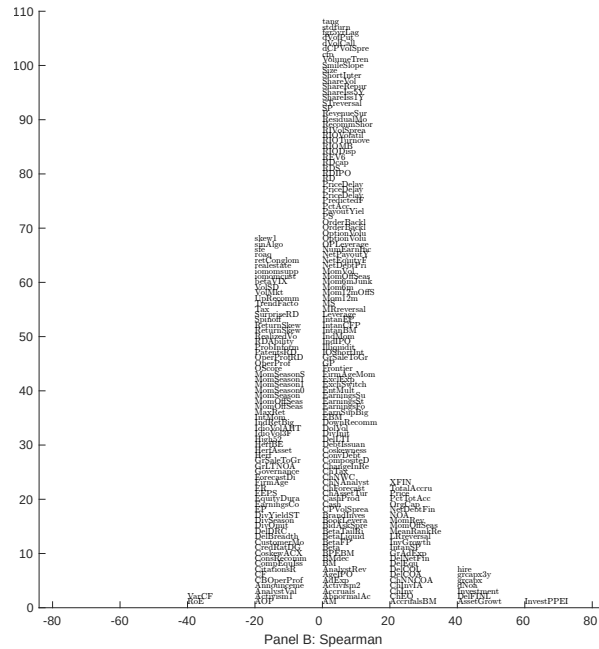
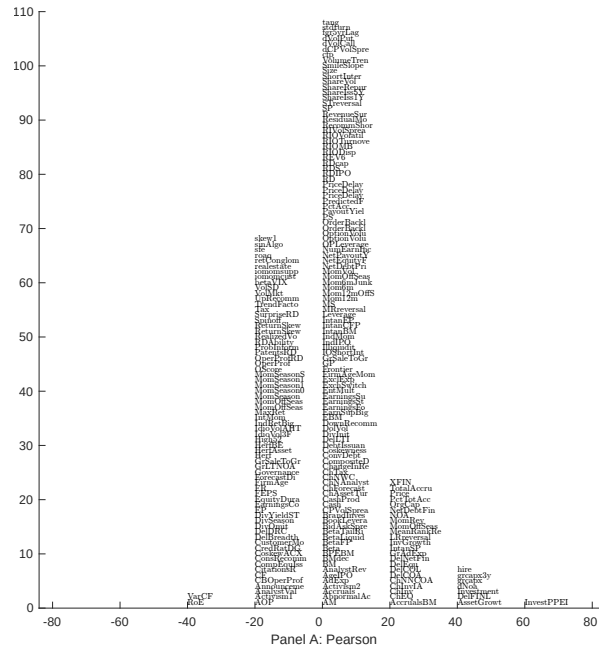


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with ALI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

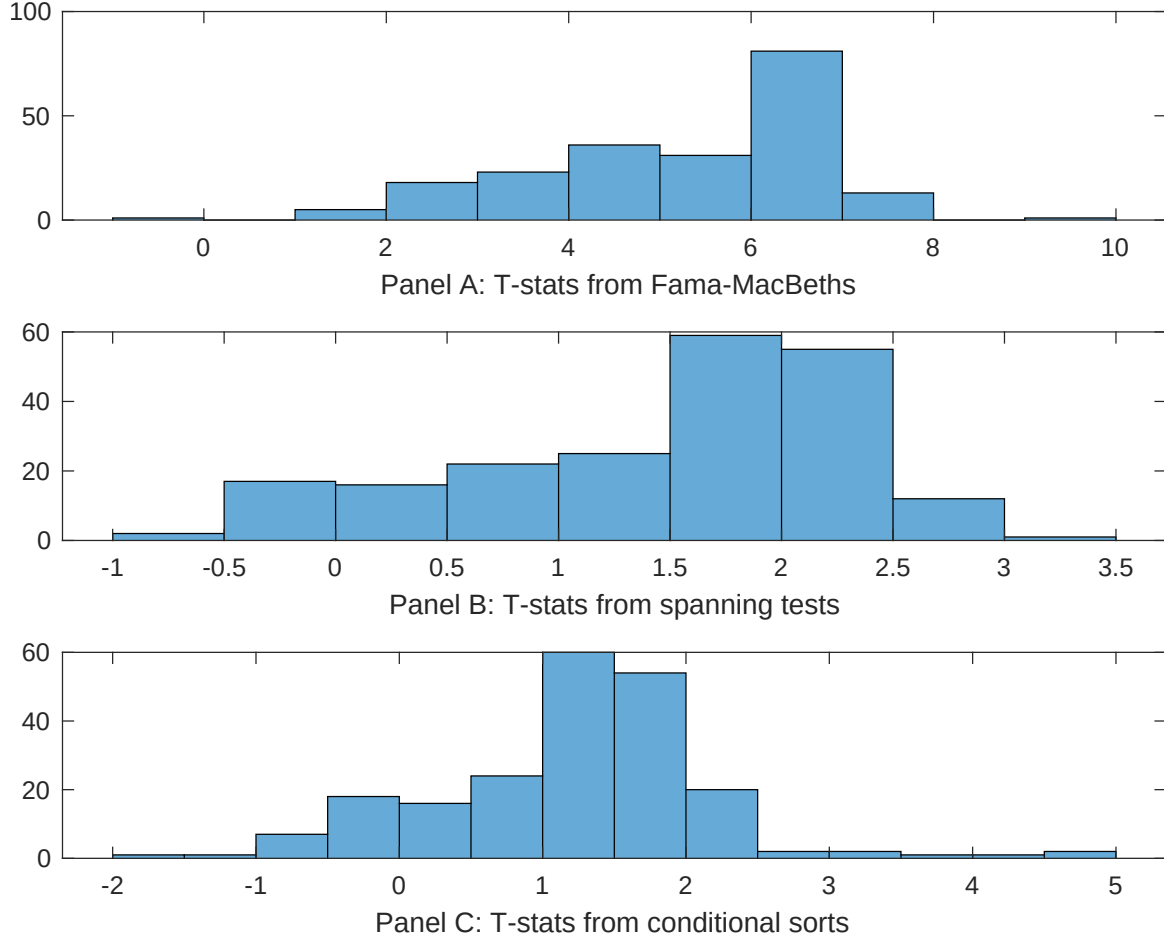


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ALI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ALI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ALI}ALI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ALI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ALI. Stocks are finally grouped into five ALI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ALI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ALI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ALI}ALI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), change in net operating assets, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [6.02]	0.14 [6.56]	0.13 [5.92]	0.13 [5.87]	0.14 [6.14]	0.13 [5.51]	0.14 [6.05]
ALI	0.33 [2.54]	0.49 [3.70]	0.64 [4.79]	0.44 [3.36]	0.40 [3.04]	0.71 [5.84]	0.25 [1.88]
Anomaly 1	0.14 [7.37]						-0.13 [-0.54]
Anomaly 2		0.12 [7.09]					0.29 [1.15]
Anomaly 3			0.55 [5.97]				0.12 [0.85]
Anomaly 4				0.11 [7.07]			0.38 [1.89]
Anomaly 5					0.93 [7.71]		0.46 [2.50]
Anomaly 6						0.80 [5.49]	0.11 [0.81]
# months	732	727	727	720	732	713	708
$\bar{R}^2(\%)$	0	0	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ALI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ALI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), change in net operating assets, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	-0.18 [-2.36]	-0.15 [-1.89]	-0.17 [-2.09]	-0.24 [-2.99]	-0.19 [-2.24]	-0.12 [-1.45]	-0.18 [-2.44]
Anomaly 1	50.95 [14.42]						39.19 [9.92]
Anomaly 2		38.64 [9.77]					23.50 [3.58]
Anomaly 3			33.98 [8.45]				-1.70 [-0.26]
Anomaly 4				37.91 [8.23]			14.01 [2.89]
Anomaly 5					17.24 [3.40]		3.75 [0.75]
Anomaly 6						25.91 [5.62]	2.28 [0.52]
mkt	10.27 [5.70]	10.32 [5.36]	10.61 [5.44]	11.08 [5.66]	10.92 [5.38]	11.21 [5.58]	10.04 [5.75]
smb	27.93 [10.70]	22.39 [7.77]	25.82 [9.02]	30.37 [10.74]	28.59 [9.65]	31.15 [10.71]	24.07 [8.93]
hml	3.32 [0.95]	1.45 [0.38]	2.69 [0.70]	5.92 [1.56]	8.37 [2.13]	4.26 [1.07]	-1.13 [-0.32]
rmw	24.82 [7.06]	22.62 [6.02]	22.08 [5.77]	25.42 [6.66]	24.87 [6.28]	25.59 [6.53]	23.92 [6.99]
cma	22.04 [3.80]	40.42 [6.93]	42.47 [7.15]	32.86 [5.03]	40.34 [4.80]	38.33 [5.46]	1.71 [0.22]
umd	1.92 [1.07]	-0.26 [-0.14]	-0.48 [-0.25]	1.22 [0.63]	2.56 [1.27]	0.57 [0.28]	0.35 [0.20]
# months	732	728	728	732	732	713	709
$\bar{R}^2(\%)$	49	42	41	40	36	38	53

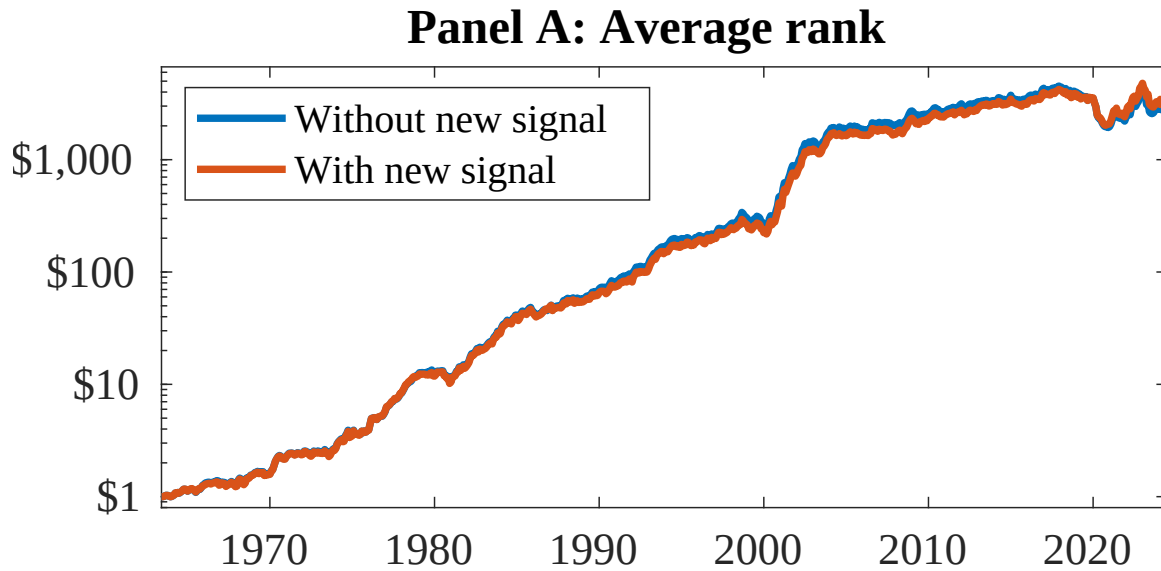


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ALI. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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